

BrewTroller PID Display

Owners Manual

v1

1. Support

If you ever have any problems or questions about your Display please email me at jason@vonnieda.org. I will respond as soon as I can.

2. Documentation

You can find the latest version of this document and other helpful information at <http://www.vonnieda.org/brewtroller-pid-display>

3. Specifications

The BrewTroller PID Display requires approximately 100mA at 5v. Do not try to use any other voltage with the Display or you may damage it.

4. Connect and Test

Attach a 4 pin female to female cable from the BrewTroller I2C header to either of the BrewTroller PID Display I2C headers. Be sure you match pin to pin. Each header is labeled SDA, SCL, +5V and GND. A straight through cable should have the same colors on each pin.

Power on your BrewTroller. Your BrewTroller PID Display should show BTPD on the top line and v1 on the bottom line for 2 seconds, then ADDR on the top line and 20 on the bottom line for 2 seconds and then go blank. If you do not see this turn off the power to your BrewTroller and recheck all of your solder joints and connections. If you still cannot get this to show please do not power up the board again and email me at jason@vonnieda.org.

5. Set Address

If you are using more than one BrewTroller PID Display in your system or you want to change the target address follow the steps below:

1. Short the two pads on the board labeled ADDR SET either from the front or the back of the board. A screwdriver or small piece of wire works well for this.
2. The Display will show ADDR and the current address of the board flashing for about 5 seconds. Remove the short if you do not want to change the address.
3. Continue to hold the pads and the address will begin to go up; about one number per second. When the address reaches your desired address just remove your short. Addresses are in hexadecimal so keep that in mind when you see addresses like 2a and 3c. Generally you should use addresses starting at 20 – 2x for however many Displays you have.
4. The Display will flash SAVE ADDR for a few seconds and the new address is saved. The change takes effect immediately.

6. Integration

When integrating the BTPD into your control panel it is important to think about current consumption. The BTPD uses about 100mA of current during normal operation. If you run the BTPD directly from your BrewTroller you can use 3 BTPDs daisy-chained before you run into any problems. If you are going to use more than 3 BTPDs you should use an external 5v power source to supply more power to the displays. Simply provide a 5v power source that can output at least 100mA per BTPD and hook it into the chain while disconnecting the power wires from the BrewTroller. See the two diagrams below to make this clear.

